Computational Analysis of Digital Communication

Course Manual

Fall 2026 (P2)



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1. Formalities and Contact

Period: P2
Language: English & Dutch
Format: Lectures, practical sessions (labs) and a supervised group project

Name	Role	Labs taught in
Assoc Prof. Dr. Philipp K. Masur	Coordinator, lecturer & lab teacher	Dutch or English
Dr. Noon Abdulqadir	Lab teacher	English
Gianni Quaedvlieg, MSc	Lab teacher	Dutch

All teachers are reached through the **Canvas inbox**. Questions about a lab go to your own lab teacher first; anything else goes to the course coordinator.

2. Course Content

This course introduces **data science** technologies and techniques to study communication processes and effects in novel and innovative ways. Learning data science and computational methods is useful because it provides new skills and much sought-after qualifications in the job market, but also allows us to tap into new areas of research and to gain a better understanding of today's society.

On the one hand, this both enables and incentivizes us to expand our **methodological toolkit**. To study political communication, we can now turn our gaze to countless communication artifacts from political parties and organizations. In media psychological research, we can make use of log data and advanced analytical procedures to measure effects at new and fine-grained levels. In the field of corporate communication, the rise of social media as central communication platforms not only requires our attention, but also enables new ways to study marketing and public relations in a natural setting and on a massive scale.

On the other hand, the digital era has also given way to new **key phenomena in communication** that are difficult to understand without a general understanding of data science technologies. Indeed, scientists are not the only ones using these technologies. More and more facets of digital life are managed at least partially by algorithms, and increasingly by large language models. Decisions of journalists, politicians, and organizations are increasingly based on metrics of collective human behavior. In the best cases, these developments can enhance lives. In the worst cases, they can foster disinformation and undermine democracy. In all cases, it is becoming increasingly difficult to understand the impact of these developments without knowing the possibilities and limits of computers.

Digital communication leaves more text behind than anyone can read: political messaging, news, social media posts, support logs, open survey answers. This course is about turning that text into numbers so a computer can analyse it, and about finding out whether those numbers measure what you think they do. Students learn how to use the statistical programming environment R to a) import text and wrangle it into a tidy format, b) represent it as tokens, word counts

and word embeddings, c) classify it with four generations of method (word frequencies, dictionaries, supervised machine learning and neural networks, and large language models such as Llama or GPT), and d) validate each of these measurements against a gold standard they coded by hand. In the last three weeks, students apply this pipeline in a research project of their own.

Based on a problem-based learning approach, each lecture introduces an empirical research problem and works towards a method that could solve it. The weeks follow the history of the field, from counting words in the 1960s to prompting a language model today. Students then implement the method themselves in two practical sessions and finish each week with a homework assignment that they start in the second practical session.

3. Learning goals

After completion of the course, you will be able to…

- identify the data-analytic problem inside a communication question, state it precisely enough that a machine could attempt it, and judge whether a computational method can answer it at all
- describe the text-classification pipeline end to end (obtain, represent, classify, validate) and name the point where each decision can go wrong
- explain what word counts, dictionaries, classic machine learning, word embeddings and large language models each cost and assume, and when the cheap method is the right one
- treat measurement as a research question in its own right, and decide when a number is not good enough to publish
- develop a research question, run a small text-analysis study in a group, and present the findings

With regard to the specific methods being taught in R, you will be able to…

- import text and wrangle it into a tidy table, and turn it into tokens, counts and embeddings
- apply dictionaries, supervised classifiers, word embeddings and large language models to the same texts
- build a gold standard, compute coder agreement (Cohen's κ, Krippendorff's α), read a confusion matrix, and report precision, recall and F1 honestly
- obtain data for a project of your own from APIs, archives and existing corpora
- make an analysis reproducible from a clean start, document the decisions behind it, and disclose what an AI tool did and where your own judgement shaped the outcome

4. Material

All lecture slides, lab materials, data, interactive explainers (the *Playground*) and the schedule are made available week by week on the course website: https://masurp.github.io/cadc/

Everything with your name on it lives on **Canvas**: announcements, readings behind a paywall, the lab folders with the course data, submissions, check-ins and grades. If it has a deadline, it

is submitted on Canvas. If a date in this manual and on the website ever differ, the website is leading.

Before the first lab, work through **Before the first lab** on the website (https://masurp.github.io/cadc/setup.html, about an hour): install R, RStudio and the course packages, and run the setup check. Before week 4, you also install Ollama and download a small language model (instructions on the week 4 page). Bring your own laptop to every lab.

5. Teaching Methods and Course Overview

The course consists of three parts. **Part I** (weeks 1 to 4) introduces four generations of text-analysis methods, one per week, and scores every one of them against texts coded by hand, including the gold standard you code yourself in week 2. Each week includes one lecture, two practical sessions (labs) in which students implement the methods introduced in the lecture using R, and one homework assignment. **Part II** (week 5) is the written exam. In **Part III** (weeks 6 to 8), students conduct a research project in small groups, supervised twice a week, and present it at a mini-conference.

- Lectures are every Monday from ***15:30 to 17:15*** in NU-2C33 (Theater 3).
- Practical sessions (labs) are every Tuesday and Thursday, from ***09:00 to 10:45*** or from ***11:00 to 12:45***. The class is split into six groups of about twenty, three in each slot. A group switches slot between the two days: groups 1 to 3 meet early on Tuesday and late on Thursday, groups 5 to 7 the other way round. Each group keeps its room on both days. Some groups are taught in Dutch and some in English (see the table below). Attendance is mandatory, and each lab opens with a short check-in.
- In weeks 6 and 7, the lab slots become supervision sessions in which your group meets its supervisor, with the same switch between Tuesday and Thursday.
- The course ends with a mini-conference on Monday 14 December 2026, 13:30 to 17:15, in HG-16A00 (Kerkzaal).

Lab group	Tuesday	Thursday	Room	Language
1	09:00 to 10:45	11:00 to 12:45	HG-05A36	Dutch
2	09:00 to 10:45	11:00 to 12:45	HG-05A37	English
3	09:00 to 10:45	11:00 to 12:45	HG-07A36	Dutch or English*
5	11:00 to 12:45	09:00 to 10:45	HG-05A36	Dutch
6	11:00 to 12:45	09:00 to 10:45	HG-05A37	English
7	11:00 to 12:45	09:00 to 10:45	HG-07A36	Dutch or English*

* Dutch or English, depending on the students in the group and their language preference.

Course Schedule

Date	Week	Type	Title	Required Readings	Assignments
	Part I: Four generations of method (weeks 1 to 4)				
26/10/26	1	**Lecture**	**Computational thinking & text as data**	Van Atteveldt & Peng (2018) *R for CSS*, ch. 2–5	
27/10/26	1	Lab A	Tidy wrangling		
29/10/26	1	Lab B	Text wrangling & exploration		Homework 1 due Sun 01/11, 23:59
02/11/26	2	**Lecture**	**Counting, dictionaries & measurement validity**	Van Atteveldt et al. (2021) Krippendorff (2019), ch. 11 *R for CSS*, text as data	
03/11/26	2	Lab A	Dictionary approaches		
05/11/26	2	Lab B	Validation lab		Homework 2 due Sun 08/11, 23:59
09/11/26	3	**Lecture**	**Learning from labels: classic ML to embeddings**	Barberá et al. (2021) Hvitfeldt & Silge (2021), ch. 6–7 Van Atteveldt et al. (2021)	
10/11/26	3	Lab A	Supervised classification		
12/11/26	3	Lab B	Word embeddings		Homework 3 due Sun 15/11, 23:59
16/11/26	4	**Lecture**	**LLMs as research instruments**	Gilardi et al. (2023) Heseltine & Clemm von Hohenberg (2024)	
17/11/26	4	Lab A	Few-shot, structured output & sensitivity		
19/11/26	4	Lab B	Validation study		Homework 4 due Sun 22/11, 23:59
	Part II: Exam (week 5)				
27/11/26	5	**Exam week**	Written exam on weeks 1 to 4 (TestVision)		Written exam 15:30 to 17:45
	Part III: Group research projects (weeks 6 to 8)				

Date	Week	Type	Title	Required Readings	Assignments
30/11/26	6	**Lecture**	**Recap & group project kickoff**	Freelon (2018) The Projects page on the website	Groups of five formed on Canvas
01/12/26	6	Supervision	Meeting with your supervisor		
03/12/26	6	Supervision	Meeting with your supervisor		Data-access plan due Sun 06/12, 23:59
07/12/26	7	No lecture			
08/12/26	7	Supervision	Meeting with your supervisor		
10/12/26	7	Supervision	Meeting with your supervisor		Poster PDF for printing, 18:00 Code and report due Sun 13/12, 23:59
14/12/26 13:30 to 17:15	8	**Mini-conference**	Presentation of the group projects		Talk and poster

6. Attendance

Attendance during the lectures is not mandatory but highly recommended: the content of the lectures (both what is on the slides and what is discussed) is the basis for the exam.

Attendance of the practical sessions is **mandatory**. Each lab opens with a ten-minute **check-in**, done on your own and without AI. Check-ins are not graded, only recorded, so a missed lab is a missed check-in. To pass the course, you need **at least six of the eight check-ins**: you may miss two labs, with a reasonable justification for your absence (illness, family or other serious reasons). Tell your lab teacher in the same week.

In weeks 6 and 7, your group is expected at every supervision session. Everyone attends the mini-conference in week 8 for the whole afternoon: you are the audience for the other groups too.

7. Literature

7.1. Suggested readings

The course draws on the following book, which is available via open access at https://cssbook.net:

- Van Atteveldt, W., Trilling, D., & Arcíla Calderón, C. (2022). *Computational Analysis of Communication: A practical introduction to the analysis of texts, networks, and images with code examples in Python and R.* Wiley.

The book is not required literature, but we recommend consulting it to dive deeper into certain topics. The following books are furthermore highly recommended, as they help you understand the content of the course in more depth. All of them are free to read online:

- Van Atteveldt, W., Trilling, D., & Arcila, C. (2026). *R for Computational Social Science.* Interactive online book that runs in the browser: https://r4css.vanatteveldt.com/
- Hvitfeldt, E., & Silge, J. (2021). *Supervised Machine Learning for Text Analysis in R.* CRC Press. Online version: https://smltar.com/
- Kuhn, M., & Silge, J. (2022). *Tidy Modeling with R.* O'Reilly. Online version: https://www.tmwr.org/
- Salganik, M. J. (2018). *Bit by Bit: Social Research in the Digital Age.* Princeton University Press. Online version: https://www.bitbybitbook.com/
- Silge, J., & Robinson, D. (2017). *Text Mining with R: A Tidy Approach.* O'Reilly. Online version: https://www.tidytextmining.com/
- Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *R for Data Science* (2nd ed.). O'Reilly. Online version: https://r4ds.hadley.nz/

7.2. Required readings

Each week, you are required to read the texts below before the lecture (see also the week pages on the website and the closing slide of each lecture). Readings behind a paywall are on Canvas.

Week 1

- Van Atteveldt, W., & Peng, T.-Q. (2018). When communication meets computation: Opportunities, challenges, and pitfalls in computational communication science. *Communication Methods and Measures, 12*(2–3), 81–92. https://doi.org/10.1080/19312458.2018.1458084
- *R for Computational Social Science*, chapters 2 to 5 (practice, so the first lab is not your first look at R).

Week 2

- Van Atteveldt, W., van der Velden, M. A. C. G., & Boukes, M. (2021). The validity of sentiment analysis: Comparing manual annotation, crowd-coding, dictionary approaches, and machine learning algorithms. *Communication Methods and Measures, 15*(2), 121–

140. https://doi.org/10.1080/19312458.2020.1869198
- Krippendorff, K. (2019). *Content Analysis: An Introduction to Its Methodology* (4th ed.), chapter 11. Sage.
- *R for Computational Social Science*, the chapter on text as data.

Week 3

- Barberá, P., Boydstun, A. E., Linn, S., McMahon, R., & Nagler, J. (2021). Automated text classification of news articles: A practical guide. *Political Analysis, 29*(1), 19–42. https://doi.org/10.1017/pan.2020.8
- Hvitfeldt, E., & Silge, J. (2021). *Supervised Machine Learning for Text Analysis in R*, chapters 6 and 7. https://smltar.com/
- Van Atteveldt et al. (2021, see week 2), the section on machine learning.

Week 4

- Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *PNAS, 120*(30), e2305016120. https://doi.org/10.1073/pnas.2305016120
- Heseltine, M., & Clemm von Hohenberg, B. (2024). Large language models as a substitute for human experts in annotating political text. *Research & Politics, 11*(1). https://doi.org/10.1177/20531680241236239

Week 6

- Freelon, D. (2018). Computational research in the post-API age. *Political Communication, 35*(4), 665–668. https://doi.org/10.1080/10584609.2018.1477506
- The Projects page on the course website, top to bottom.

8. Evaluation

The evaluation of the course is based on three graded components: a written exam, four individual homework assignments, and a group project.

Component	When	Weight
Written exam	Week 5 · individual · no AI	40%
Homework assignments	Weeks 1 to 4 · individual · four assignments	30%
Group project	Weeks 6 to 8 · one grade per group	30%

In addition, two things are required to pass, without a grade: **at least six of the eight check-ins** (see section 6) and **all four homework assignments** handed in. A homework that is not handed in cannot be averaged away.

Please note: in order to pass the course, all three components need to be passed. You cannot compensate for a failed component with another one.

Language. Anything you write out yourself (homework code comments and reflections, answers to open exam questions, the project report) may be written in English or in Dutch. The multiple-choice questions in the exam and the check-ins, and all course materials, are in English. The talk and the poster at the mini-conference must be in English.

8.1. Exam

After the first four weeks, there is a written exam (**40% of the final grade**):

- The exam takes place on Friday 27 November 2026, 15:30 to 17:45, on TestVision. Students with an approved extra time arrangement finish at 18:15. The room will be announced on Canvas.
- It consists of multiple-choice and open questions on everything from weeks 1 to 4: lecture content (both what is on the slides and what is discussed), the lab materials, and the required readings.
- The exam tests interpretation, not typing: you read R output and say what it means, read code you are shown, find the flaw in a pipeline, and judge whether a measurement can be believed. You are never asked to write code from scratch.
- Many questions use the data, code and traps from the labs. The check-ins are your chance to practise similar questions in the room.
- The exam is taken without AI and without notes.
- Please register for the exam via VUweb.

8.2. Homework assignments

In each of the first four weeks, you hand in an individual homework assignment (**30% of the final grade**). Each homework is Part 2 of that week's Thursday lab (Lab B): you start it in the room, with your teacher around, and finish it at home.

No.	Homework	What you do	Due
1	**New Year's resolutions**	Clean, count and compare the words in 5,000 resolution tweets, and find out where counting words misleads you.	Sun 01/11/26 23:59
2	**Is the dictionary right?**	The Bing dictionary against the gold standard you coded with your partner: confusion matrix, precision, recall, and the tweets it gets wrong.	Sun 08/11/26 23:59
3	**Word counts or embeddings?**	A classifier that finds one category of resolution tweets, trained on word counts and on embeddings, and what accuracy hides.	Sun 15/11/26 23:59
4	**Your own measurement**	Your week 2 codebook as the prompt for a language model, on your own gold standard: which measurement would you defend?	Sun 22/11/26 23:59

- The lab folder for Lab B contains a Quarto template for the homework, with every step as a heading. The exact task is in Part 2 of that week's Lab B.

- Each homework contains (A) your code for every step, run and rendered; (B) three short reflections of about 100 words each on **your own results**, quoting specific texts and numbers; and (C) an AI disclosure (see section 8.4), which is also required if you used no AI at all.
- You hand in **one rendered HTML file** on Canvas, by Sunday 23:59 of the same week. A file that does not render cannot be graded. Homework 2 and 4 use the gold standard you code with a partner in week 2: keep that file safe.
- The homework is graded by your lab teacher, with a few lines of feedback, on the following scale. Your homework grade is the arithmetic mean of the four grades, not rounded.

Grade	What it looks like
Excellent (10)	Everything below, and the reflections explain *why* something happened in your results, not only *that* it happened
Good (8)	The file renders, every step is done, the reflections refer to your own output (specific texts, specific numbers), and the AI disclosure is complete
Sufficient (6)	The file renders and the steps are done, but the reflections stay general, or a step is incomplete
Insufficient (4)	The file does not render, steps or reflections are missing, or there is no AI disclosure

8.3. Group project

In weeks 6 to 8, you conduct your own study in a group: a question you chose, data you obtained yourselves, and a measurement you validated. The project is presented at the mini-conference and graded per group (**30% of the final grade**).

- **Groups of five** are formed on Canvas in the week 6 lecture. You may cross lab groups, but a group is supervised in one lab slot. Your lab teacher is your supervisor; each group meets its supervisor in both sessions of weeks 6 and 7.
- **Week 6:** form a group, pick a question, and get data in hand. Hand in a one-page **data-access plan** on Canvas (source, how you get it, legal basis, what you will have, your fallback). The Projects page on the website lists which data sources are realistic in three weeks.
- **Week 7:** build the measurement, hand-code a sample, compute agreement, and evaluate the automated measure against it. **A project without a validation section cannot pass.**
- **Week 8:** a ten-minute talk per group plus five minutes of questions at the mini-conference on Monday 14 December 2026, and an A0 research poster shown during the poster gallery. The poster PDF is due on Canvas by Thursday 10 December 18:00 if you want it printed for you; later means you print it yourselves. Code and rendered report are due on Sunday 13 December 23:59.
- Anything involving participants (data donation, interviews, surveys) needs ethics approval before you start. API keys, tokens and raw personal data never go into your shared

code.

The project is graded on the following criteria. The criteria are fixed; the weights may still change before week 6.

Criterion	What we look for	Weight
Question & framing	A communication question that a computational method can actually answer, stated precisely, with the construct defined before the tool is chosen	20%
Data & access	A defensible source, an honest account of what it covers and what it misses, sampling decisions made on purpose and documented	20%
Measurement & validation	A gold standard you built, agreement reported, the automated measure evaluated against it, and the right metric for the class balance you have	30%
Interpretation	Conclusions that match the evidence, limitations named precisely, and a clear statement of which number you would defend	20%
Presentation & poster	Ten minutes that land, a poster that reads at two metres, and answers that hold up under questioning	10%

There is one grade per group. If the work in your group is genuinely unequal, tell your supervisor **before** week 8; afterwards it is too late to adjust anything.

8.4. Use of AI

The course runs two lanes on purpose, and always tells you which one you are in. **Part 1 of every lab, the check-ins and the exam** are done without AI, including code completion tools such as GitHub Copilot: this is where you build, and where we find out, what you can do yourself (reading output, reading code, judging a design), which is the capability that using these tools well depends on. In **Part 2 of every lab, the homework and the project**, AI tools are permitted, as long as you disclose how you used them and can account for everything you hand in.

We distinguish four kinds of use, by what the tool did to the work:

Category	What it means	What we ask of you
Instrument	The model performs a defined task whose output enters your work: classifying your texts, generating code you run, producing labels you analyse.	Method-level disclosure: model and version, how the prompt was designed, and validation against your own judgement.
Assistant	Editing, grammar, translation, debugging, help searching. The intellectual content is yours; the tool touched the surface.	Which tool, for what, and what you did with its output: what you kept, changed and rejected.
Thinking partner	The model works on your own notes and drafts, helping you stress-test ideas that are already yours.	What you discussed and what came out of it. You can defend every claim without the output in front of you.
Surrogate	The model produces the work and you put your name on it.	Not acceptable. This falls under the VU fraud and plagiarism regulation, and no disclosure makes it acceptable.

Every homework ends with an **AI disclosure**: one block for each category you used, answering four questions: (1) which tool, and when; (2) what for: which step, and what you asked it; (3) what you did with it: what you kept, what you changed, and at least one thing you rejected, with the reason; (4) how you checked that the code or claim is right. The standard behind it is that you must be able to reconstruct and defend the reasoning without the output in front of you. A disclosure such as "ChatGPT was used for help with the code" counts as missing. Templates and worked examples are on the AI use page of the website.

8.5. Deadlines, late work and resits

- **Late work.** If you cannot meet a deadline, message the coordinator on Canvas **before** the deadline, not after. Without prior agreement, a late homework is graded Insufficient.
- **An insufficient homework.** An Insufficient homework is not redone: the homework grade is the mean of all four, so one or two Insufficient grades can be compensated by the others. Three Insufficient grades will most likely bring the mean below 5.5, and with it the course is failed.
- **Resit.** Only the exam has a resit, in the resit exam week in January/February. The homework and the group project are coursework and cannot be resat: a failed group project means the course is failed.
- **Final grade.** The final grade is the weighted average of the three components: exam 40%, homework 30%, group project 30%. Each component must be passed with a grade of 5.5 or higher; the homework mean is not rounded for this. The final grade is rounded to the nearest half point (an 8.3 becomes 8.5, an 8.2 becomes 8.0), and final grades between 5.0 and 6.0 are not given: 5.5 or higher rounds up to 6.0, anything below 5.5 rounds down to 5.0.
- **Feedback and questions about a grade.** Check-ins are discussed in the room; each homework gets a grade and a few lines of feedback; the project gets its feedback at the

conference. Questions about a grade go to your lab teacher first and the coordinator second. The formal route is the Examination Board, and it stays open to you.

The Teaching and Examination Regulations (OER) of the programme apply in full.

9. Getting Help

In this course, you learn how to use R to conduct computational analyses. Most problems are solved fastest in the lab: bring them there.

- **Before the first lab**, work through the setup page on the website and run the setup check. If something does not install, bring the output of the check to the first lab, or paste it into your message when you ask for help.
- **Questions about a lab or a homework** go to your lab teacher, in the lab if at all possible: they get answered fastest there, and help others at the same time.
- **Everything else** goes to the course coordinator, via the Canvas inbox.

When you ask for help with code, say what you ran, what you expected, and paste the full error message.